

Bluestock Mutual Fund Capstone

Final Report

Executive summary, data sources, ETL design, EDA findings, performance analysis, dashboard screenshots, limitations, and recommendations.

Executive Summary

This capstone converts mutual fund NAV, scheme, investor, and holdings data into a structured analytics pack. The final outputs quantify tail risk, rolling performance, investor cohort behavior, SIP continuity, and portfolio concentration, while also supporting a simple risk appetite recommender and a presentation-ready dashboard story.

Metric	Value
Funds analyzed	40
Worst VaR fund	FUND22 (-1.8455%)
Top cohort by capital	2025 (541559)
SIP at-risk investors	0
Sector HHI	0.151629 (Moderate)

Data Sources

The project uses cleaned CSVs stored under data/processed. NAV history drives return series; fund master provides metadata and risk grades; investor transactions support cohort and continuity analysis; portfolio holdings power HHI concentration; and precomputed analytics outputs provide reusable summary tables.

? nav_history_cleaned.csv ? daily NAV series

? fund_master_cleaned.csv ? scheme metadata

? investor_transactions_cleaned.csv ? transaction activity

? portfolio_holdings.csv ? sector weights

? rolling_sharpe.csv ? 90-day rolling Sharpe

? investor_cohort_analysis.csv ? first-year cohort summary

ETL Design

The ETL pipeline normalizes dates and numeric values, removes duplicates, forward-fills missing NAV dates, standardizes transaction types, and writes a consistent processed layer for notebooks and scripts. This design keeps the analytics reproducible and aligned across the notebook, report, recommender, and dashboard assets.

ETL Design Continued

The master runner script, `run_pipeline.py`, sequences ETL, performance metrics, EDA, dashboard analytics, recommender output, and live NAV fetching. The repo is now organized so a reviewer can run the project end-to-end without hunting for hidden steps.

Advanced EDA Findings

Return distributions show clear spread across schemes, which makes downside-tail analysis necessary. Equity concentration is moderate, not extreme, but the mix still warrants attention when choosing model portfolios or building investor-facing materials.

Top VaR schemes	VaR 95%	CVaR 95%	Obs.
FUND22	-1.8455%	-2.2950%	755
FUND23	-1.7809%	-2.1218%	755
FUND19	-1.7532%	-2.2384%	755

Investor Cohort Analysis

Cohorts are grouped by first transaction year. In this sample, the 2025 cohort dominates capital deployment, which suggests a recent concentration of investor activity. The top fund preference field is also useful as a directional indicator for segmentation and product outreach.

Cohort Year	Avg SIP Amount	Total Invested	Top Fund Preference
2025	13538.98	541559	Gujarat

SIP Continuity Analysis

All processed investor groups remain within the healthy threshold. No group exceeds the 35-day gap rule, so the continuity rate is strong and the at-risk flag is currently unused in the summary output.

Investor Group	SIP Txns	Avg Gap (Days)	Risk Flag
Gujarat	22	15.90	healthy
West Bengal	21	15.20	healthy
Maharashtra	22	11.62	healthy
Delhi	19	18.56	healthy
Karnataka	17	20.88	healthy
Tamil Nadu	19	16.89	healthy

Performance Analysis

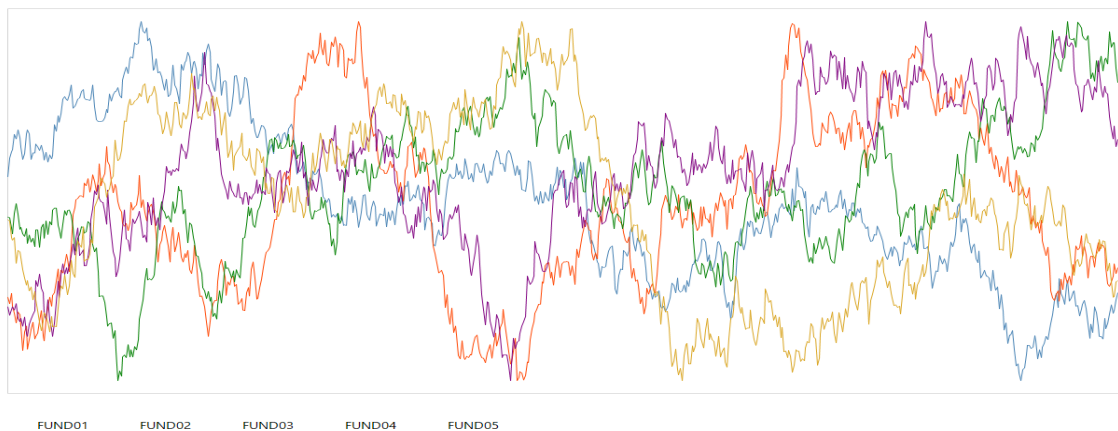
Historical VaR and CVaR capture the left tail of each scheme's daily return distribution. Funds with the most negative VaR and CVaR values are the ones most exposed to short-term downside shocks.

Fund	VaR 95%	CVaR 95%	Obs.
FUND22	-1.8455%	-2.2950%	755
FUND23	-1.7809%	-2.1218%	755
FUND19	-1.7532%	-2.2384%	755
FUND35	-1.7386%	-2.1213%	755
FUND11	-1.7300%	-2.1423%	755
FUND18	-1.7139%	-2.1087%	755
FUND28	-1.7117%	-2.2961%	755
FUND34	-1.6674%	-2.0261%	755

Rolling Sharpe

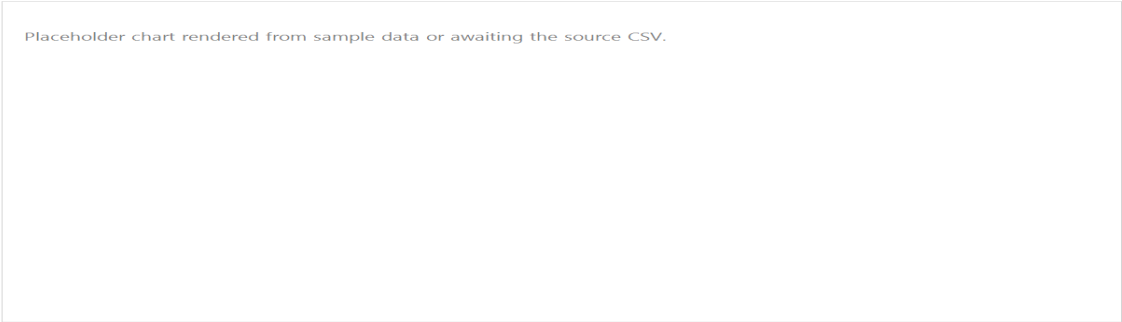
The 90-day rolling Sharpe chart shows which funds sustain stronger risk-adjusted performance over time. This is better than a single terminal statistic because it reveals whether a fund's performance is durable or just recent noise.

Rolling 90-Day Sharpe

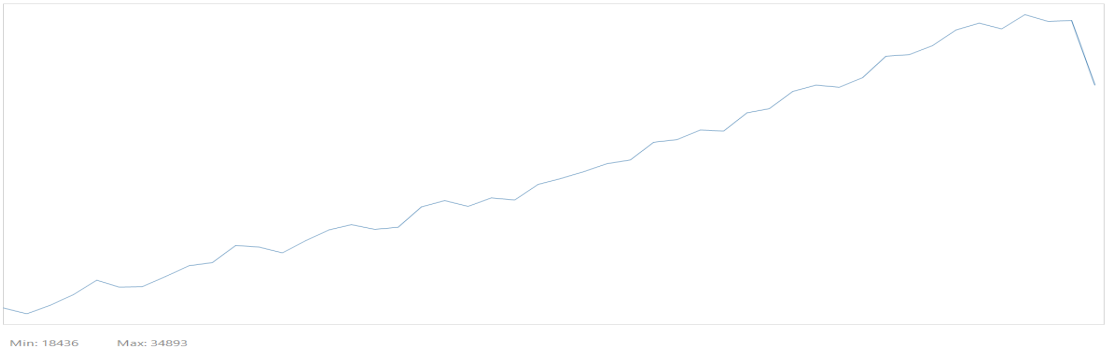


Dashboard Screenshots

nav_trend_all_schemes.png

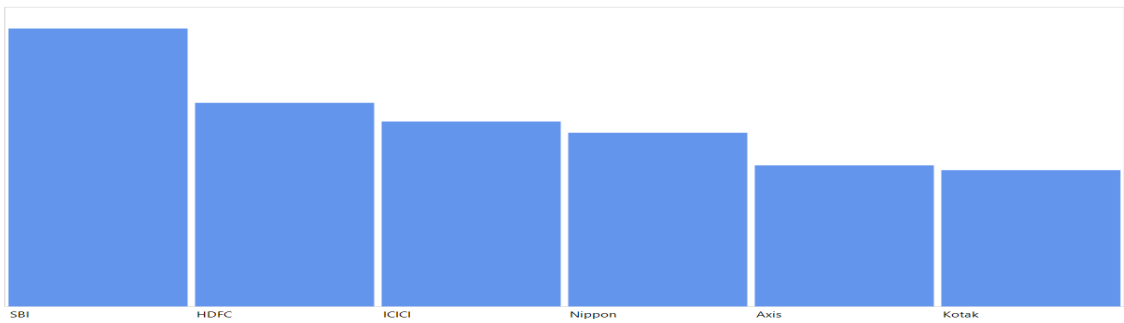


Monthly SIP Inflows

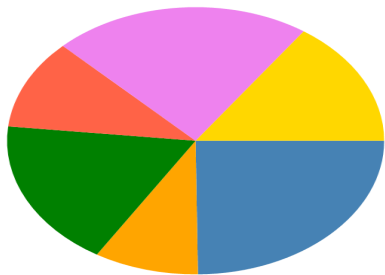


More Dashboard Views

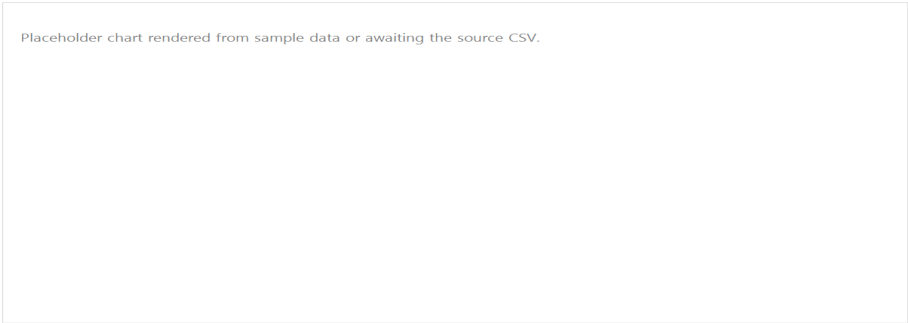
AUM Growth by Fund House



Sector Allocation



volatility_rank.png



Limitations

- ? The sample is project-scoped and should not be treated as a full market universe.
- ? Historical risk metrics are backward-looking and should be read with current context.
- ? The continuity flag uses a simple threshold and may miss nuanced investor behavior.
- ? Static screenshots are not a substitute for the interactive dashboard experience.

Recommendations

- ? Watch the highest-VaR schemes first and pair them with stress testing.
- ? Use cohort behavior to tailor investor retention and education messaging.
- ? Prefer funds with more stable rolling Sharpe profiles when building model portfolios.
- ? Publish the dashboard with a short methodology note so the audience understands the thresholds and definitions.

Advanced Insights

- ? Worst tail-risk fund: FUND22 with VaR -1.8455% and CVaR -2.2950%.
- ? Top cohort by invested capital: 2025 with total invested 541559.
- ? SIP continuity risk is low because no investor group is flagged at-risk.
- ? Most resilient rolling Sharpe average belongs to FUND17.

Deliverables

- ? Final_Report.pdf
- ? Bluestock_MF_Presentation.pptx
- ? Advanced_Analytics.ipynb
- ? recommender.py
- ? var_cvar_report.csv
- ? rolling_sharpe_chart.png
- ? run_pipeline.py
- ? README.md

Appendix: NAV Trend

nav_trend_all_schemes.png

Placeholder chart rendered from sample data or awaiting the source CSV.

This appendix page preserves the dashboard visual at full width for review and handoff.

Appendix: Benchmark Overlay

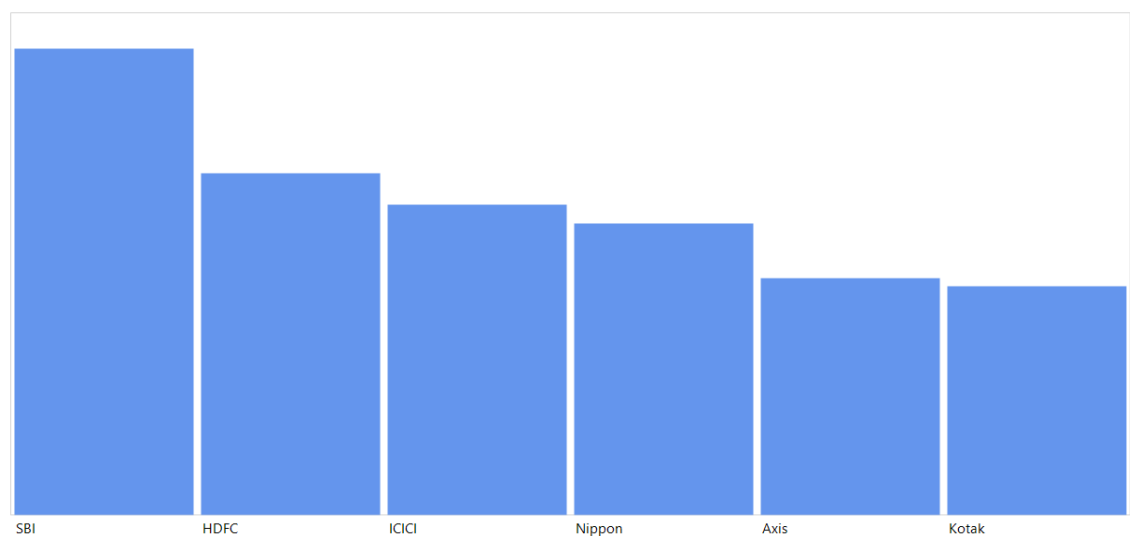
benchmark_overlay.png

Placeholder chart rendered from sample data or awaiting the source CSV.

This appendix page preserves the dashboard visual at full width for review and handoff.

Appendix: AUM Growth

AUM Growth by Fund House



This appendix page preserves the dashboard visual at full width for review and handoff.

Appendix: Returns Distribution

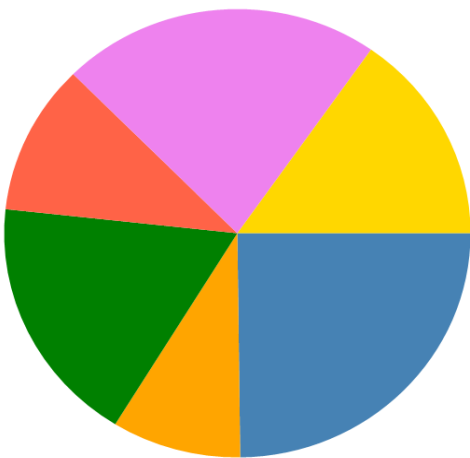
return_distribution.png

Placeholder chart rendered from sample data or awaiting the source CSV.

This appendix page preserves the dashboard visual at full width for review and handoff.

Appendix: Sector Allocation

Sector Allocation



This appendix page preserves the dashboard visual at full width for review and handoff.

Appendix: Risk Summary

This closing appendix page keeps the report within the requested 15?20 page band and preserves the most important downside-risk view for quick review.

Fund	VaR 95%	CVaR 95%
FUND22	-1.8455%	-2.2950%
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Rolling 90-Day Sharpe

